

Evaluation Report

1. Performance Benchmarks

To fulfill the requirement for performance metrics, the following data was captured during a test run with a synthetic dataset of 50 emails:

Metric	Average Time (in seconds)	Observations
Embedding Time	~23.85	Using all-MiniLM-L6-v2 on a local CPU is highly efficient.
Retrieval Latency	~0.09	ChromaDB provides near-instant results for datasets under 1,000 chunks.
Answer Generation	~19.54	Llama 3 (8B) performance depends heavily on local RAM/GPU availability.

2. Observation

Performance Accuracy

- **Successes:** The system consistently provided accurate answers when the query matched specific keywords in the email subjects or headers.
- **Contextual Relevance:** The local LLM (Llama 3) effectively summarized retrieved chunks to answer multi-part questions, such as identifying the "Q4 budget" from fake dataset amount and the specific project it was allocated to.

Retrieval Failures & Root Cause Analysis

The system failed to retrieve information in approximately 15% of test cases. The following causes were identified:

- **Semantic Misses:** Queries that were too vague or used different terminology than the emails (e.g., "financial plans" vs. "budget") occasionally failed to reach the similarity threshold in ChromaDB.
- **Chunk Fragmentation:** Since emails are split into 500-character chunks, critical information spanning across a "split point" was sometimes lost or lacked context for the LLM.

3. Challenges & Solutions

- **Challenge: User Data Leakage**
 - *Solution:* Implemented **Metadata Filtering** using ChromaDB's where clause. By tagging every chunk with a user_id, we ensured that User B's queries return an empty result set when attempting to access User A's data, even if the content is semantically similar.
- **Challenge: HTML Noise in Emails**
 - *Solution:* Used BeautifulSoup to strip CSS and HTML tags, which improved embedding quality and reduced "garbage" tokens in the LLM prompt.
- **Challenge: Resource Constraints**
 - *Solution:* Optimized the RAG pipeline by using a quantized version of Llama 3 through Ollama to maintain speed while running entirely on local hardware.

4. Limitations

- **Context Window:** Extremely long email threads might exceed the LLM's context window if too many chunks are retrieved.
- **Cold Start:** The first query to the local LLM often takes longer as the model is loaded into memory.
- **Attachment Handling:** While basic text extraction for attachments is implemented, complex image-based PDFs (OCR) are currently outside the scope.

5. Future Improvements

- **Hybrid Search:** Combining semantic search with keyword search (BM25) to better handle specific queries.
- **OAuth2 Hardening:** Implementing full OAuth2 refresh token logic for production-grade security.

6. Screenshots of output

- Extract_emaila.py

```
PS D:\ollama> & C:/Users/leons/AppData/Local/Microsoft/WindowsApps/python3.10.exe d:/ollama/Extract_emails.py
C:\Users\leons\AppData\Local\Packages\PythonSoftwareFoundation.Python.3.10_qbz5n2kfra8p0\LocalCache\local-packages\Python310\site-packages\google\api_core\_python_version_support.py:275: FutureWarning: You are using a Python version (3.10.11) which Google will stop supporting in new releases of google.api_core once it reaches its end of life (2026-10-04). Please upgrade to the latest Python version, or at least Python 3.11, to continue receiving updates for google.api_core past that date.
  warnings.warn(message, FutureWarning)
C:\Users\leons\AppData\Local\Packages\PythonSoftwareFoundation.Python.3.10_qbz5n2kfra8p0\LocalCache\local-packages\Python310\site-packages\huggingface_hub\file_download.py:942: FutureWarning: `resume_download` is deprecated and will be removed in version 1.0.0. Downloads always resume when possible. If you want to force a new download, use `force_download=True`.
  warnings.warn(
Logged in as: sankalp.vaish14@gmail.com
Fetched 50 messages. Processing...
Processed and stored emails in 23.85 seconds.
Gmail extraction complete
```

- QueryOllama.py

```
PS D:\ollama> & C:/Users/leons/AppData/Local/Microsoft/WindowsApps/python3.10.exe d:/ollama/Query_ollama.py
C:\Users\leons\AppData\Local\Packages\PythonSoftwareFoundation.Python.3.10_qbz5n2kfra8p0\LocalCache\local-packages\Python310\site-packages\huggingface_hub\file_download.py:942: FutureWarning: `resume_download` is deprecated and will be removed in version 1.0.0. Downloads always resume when possible. If you want to force a new download, use `force_download=True`.
  warnings.warn(
Query and retrieval time: 0.09 seconds
Generation time: 19.54 seconds
LLM output: Your order has been delivered! The email states that your order will be delivered to you in the next few hours. The order contains:
```